

## ME23312 ENGINEERING THERMODYNAMICS

Unit I – Basics, Zeroth and First Law · Presentation 1 of 10

# Introduction and Thermodynamic Systems

What thermodynamics is about · macroscopic and microscopic views · the continuum · system, boundary and surroundings · closed, open and isolated systems

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3. Closed, open and isolated systems
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# 1

## What thermodynamics is about

Energy, its transformations and its limits

# Thermodynamics – the science of energy

- Greek *therme* (heat) + *dynamis* (power): born from the study of the steam engine
- **First law** – energy is conserved
- **Second law** – the direction of processes and the limit on converting heat into work
- Governs power plants, IC engines, gas turbines, refrigerators, air conditioners and chemical plants

**KEY POINT** Every device that produces power, provides refrigeration or processes a fluid obeys these two laws.

# What Unit I covers

- Systems, properties and processes; thermodynamic equilibrium
- Displacement work and the  $p$ – $V$  diagram
- Thermal equilibrium, the Zeroth law, temperature and temperature scales
- The first law for closed and open systems – steady and unsteady flow

**KEY POINT** Course outcome CO1: apply the first law of thermodynamics to closed and open systems for thermal equipment.

# Two points of view

## Macroscopic (classical)

- Bulk properties:  $p$ ,  $V$ ,  $T$
- Few variables, directly measurable
- No assumption about molecular structure
- The approach of this course

## Microscopic (statistical)

- Matter = a very large number of molecules
- Statistical averaging of molecular behaviour
- Kinetic theory, statistical thermodynamics
- Same results for measurable quantities

# The continuum assumption

- Matter treated as continuous – no voids
- Density and pressure defined at every point as limits of mass/volume and force/area
- Valid when the smallest volume of interest holds a very large number of molecules – mean free path  $\ll$  system size
- Fails only for highly rarefied gases (high vacuum, upper atmosphere)

# Where thermodynamics is used

## Power generation

Steam and gas turbine plants convert the chemical energy of fuel into electricity; every plant efficiency is limited by the laws of thermodynamics.

## IC engines

Otto and Diesel cycles, mean effective pressure and fuel economy are first- and second-law calculations.

## Refrigeration and air conditioning

Pumping heat from a cold space to warm surroundings – within the limits set by the second law.

## Flow machinery

Turbines, compressors, pumps, nozzles, boilers and condensers are analysed with the steady flow energy equation.



# 2

## Thermodynamic systems

System, boundary, surroundings and the three types of system

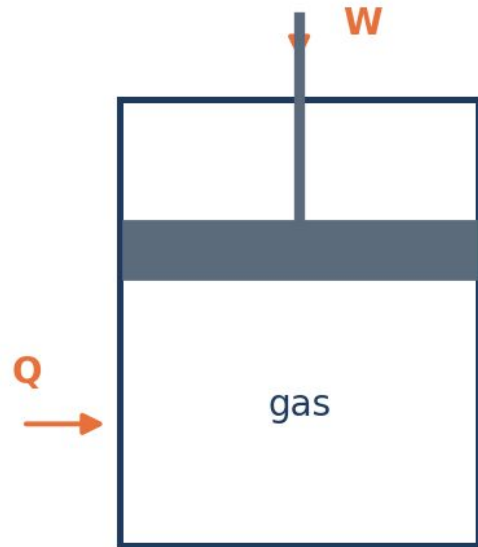
# System, boundary and surroundings

- **System** – the quantity of matter or the region in space chosen for study
- **Surroundings** – everything outside the system
- **Boundary** – real or imaginary surface of zero thickness; fixed (rigid tank) or movable (piston face)
- System + surroundings = **universe**

**KEY POINT** Draw the system and mark its boundary before writing any equation – the boundary decides which heat, work and mass transfers count.

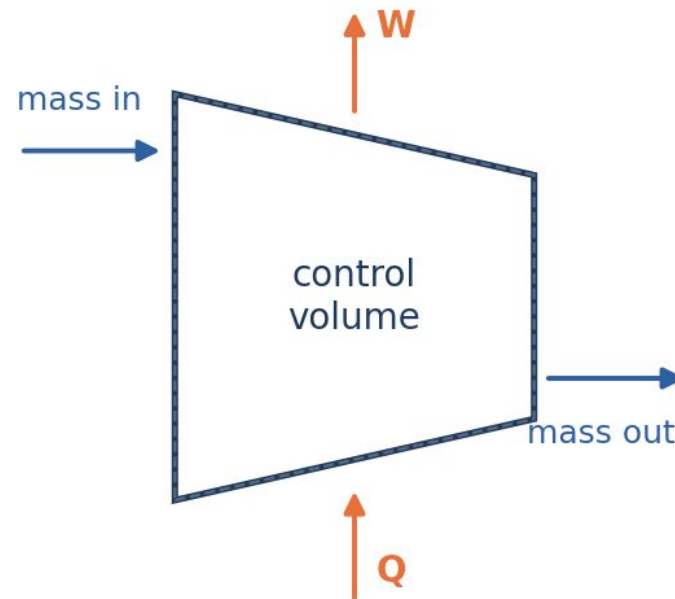
# Closed, open and isolated systems

**Closed system (control mass)**

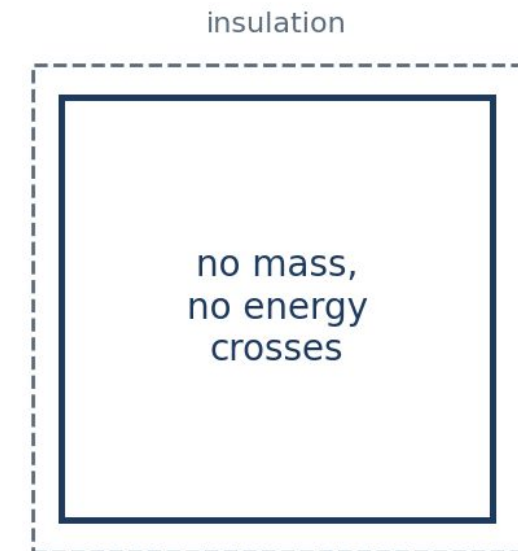


mass fixed; heat and work cross

**Open system (control volume)**



**Isolated system**



Closed system (control mass) · open system (control volume) · isolated system

# The three types of system

| Type            | Mass crosses? | Energy crosses?        | Also called    | Examples                                                      |
|-----------------|---------------|------------------------|----------------|---------------------------------------------------------------|
| <b>Closed</b>   | No            | Yes – heat and/or work | Control mass   | Gas in a piston–cylinder, sealed pressure cooker, car battery |
| <b>Open</b>     | Yes           | Yes                    | Control volume | Turbine, compressor, boiler, nozzle, pump, water heater       |
| <b>Isolated</b> | No            | No                     | —              | Insulated rigid flask; the universe                           |

# The three systems in more detail

- **Closed system** – fixed mass; the boundary may move (gas pushing a piston) but no matter enters or leaves
- **Open system** – analysed as a **control volume**, a fixed region enclosed by a **control surface** through which mass flows carrying its energy; almost all practical machines
- **Isolated system** – exchanges neither mass nor energy, so its total energy is constant; an idealisation approached by a well-insulated rigid sealed vessel

# Homogeneous and heterogeneous systems

- **Phase** – matter homogeneous throughout in chemical composition and physical structure (solid, liquid or gas)
- **Homogeneous system** – a single phase: air in a room, water in a tank
- **Heterogeneous system** – two or more phases: water + steam in a boiler drum, ice + water in a glass

# Summary

- Thermodynamics = science of energy; first law (conservation) and second law (direction and limits)
- Macroscopic approach: few measurable bulk properties; no molecular model needed
- Continuum: properties defined at every point; fails only for rarefied gases
- System, boundary, surroundings; system + surroundings = universe
- Closed – fixed mass; open – control volume with mass flow; isolated – no mass, no energy
- Homogeneous = one phase; heterogeneous = two or more phases

# Next: Properties, state, process and cycle

- Intensive and extensive properties; specific properties
- State and the two-property rule; process, path and cycle
- Point functions and path functions